



**SYDNEY GIRLS HIGH SCHOOL
TRIAL HIGHER SCHOOL CERTIFICATE**

2000

MATHEMATICS

2 UNIT

**Time Allowed – 3 hours
(Plus 5 minutes reading time)**

DIRECTIONS TO CANDIDATES NAME

- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2000 HSC Examination Paper on this subject

QUESTION 1

- (a) Find, correct to two decimal places, the value of: [2]

$$\frac{9.65^2 - 3.43^3}{\sqrt{56.5}}$$

- (b) Find the values of a and b given: [2]

$$\frac{2}{3 - \sqrt{2}} = a + b \sqrt{2}$$

- (c) Factorize the equation and solve for x [2]
 $3x^2 - 2x - 5 = 0$

- (d) Solve for x and mark the set of values on a number line: [2]
 $|2x - 3| \leq 7$

- (e) Express $0.4\dot{7}$ as an exact fraction in its simplest form. [2]

- (f) Given $\frac{1}{P} = \frac{1}{Q} + \frac{1}{R}$

(i) make Q the subject of the formula

(ii) Hence, or otherwise, find Q when $P = 0.2$ and $R = 0.5$ [2]

QUESTION 2

- (a) On a number plane, mark the origin O and the points A (4,1), B (6,3) and C (2,7). Join A to B, B to C and C to A [1]
- (b) Find the gradient of line AB [1]
- (c) Find the equation of line AB [1]
- (d) Write down the acute angle (to the nearest degree) line AB makes with the x -axis. [1]
- (e) Find, in exact form, distance AB [1]
- (f) Prove $\triangle ABC$ is right-angled [2]
- (g) Find the mid-point of AC [1]
- (h) Using part (g), or otherwise, find the co-ordinates of D which would form the rectangle ABCD. (Show all necessary working) [1]
- (i) Find the exact area of rectangle ABCD [1]
- (j) Find the shortest (perpendicular) distance from the origin to line AB (leave your answer in exact form). [2]

QUESTION 3

(a) Differentiate:

(i) $4x^3 - \frac{4}{x^3}$ [1]

(ii) $\ln\left(\frac{x^2-5}{x+3}\right)$ [2]

(iii) $x^2 \cdot e^{\cos x}$ [2]

(b) Find: (i) $\int \frac{dx}{3x-1}$ [1]

(ii) $\int (2x+1)^9 dx$ [1]

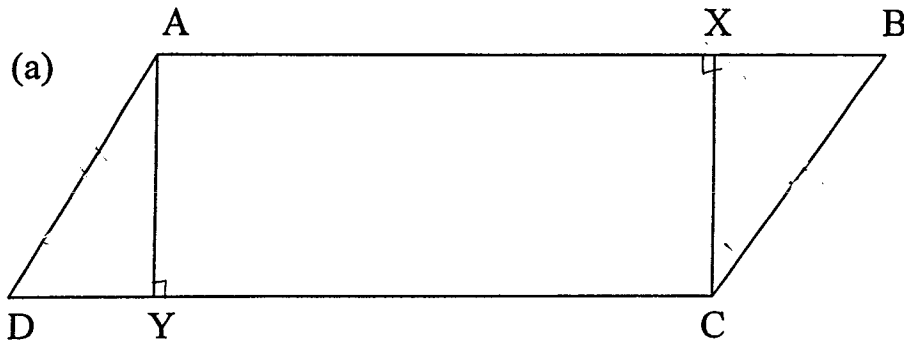
(c) Evaluate, leaving your answer in exact form

$\int_0^1 e^{4x} dx$ [2]

(d) (i) If $y = x \cdot \sin x + \cos x$ find $\frac{dy}{dx}$ [1]

(ii) Using part (i), or otherwise, find the value of: $\int_{\frac{\pi}{2}}^{\pi} x \cdot \cos x \cdot dx$ [2]

QUESTION 4



ABCD is a parallelogram
 $CX \perp AB$ and $AY \perp DC$

- (i) Prove $\triangle ADY$ and $\triangle CBX$ are congruent
- (ii) Hence, or otherwise, prove $AY = CX$

[4]

- (b) A ship leaves Port A and travels 55km on bearing 067° to reach Port B. It then travels on bearing 310° for 25km until it reaches Port C (see figure below).

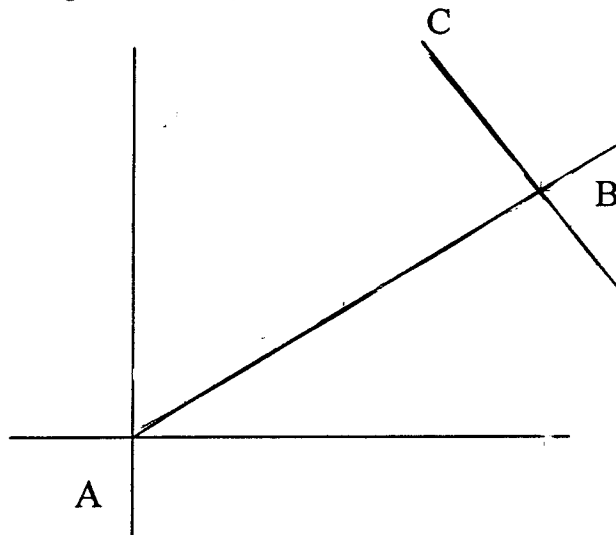


Figure not to scale

[3]

- (i) Find $\angle ABC$
- (ii) Find distance AC (to the nearest whole km)

- (c) A parabola has its focus at (3,3) and its directrix is the line $y = -1$

- (i) Sketch the parabola showing the vertex, focus and directrix
- (ii) Write down the equation of the axis of symmetry
- (iii) Write down the equation of the parabola
- (iv) Find the equation of the tangent to the parabola at the point where the parabola intersects the y -axis.

[5]

QUESTION 5

- (a) Consider the curve given by $y = 3x^2 - x^3$
- (i) Find the stationary points and determine their nature.
 - (ii) Sketch the curve, indicating all essential features.
 - (iii) Find the equation of the normal to the curve where the curve crosses the x – axis
- [7]
- (b) In an arithmetic progression the sum of the 3rd and 4th terms is 24. Further, the sum of the 5th and 6th terms is 32. Find.
- (i) the first three terms
 - (ii) the sum of the first 20 terms.
- [3]
- (c) For what values of m has the equation $4x^2 + (1+m)x + 1 = 0$ equal roots.
- [2]

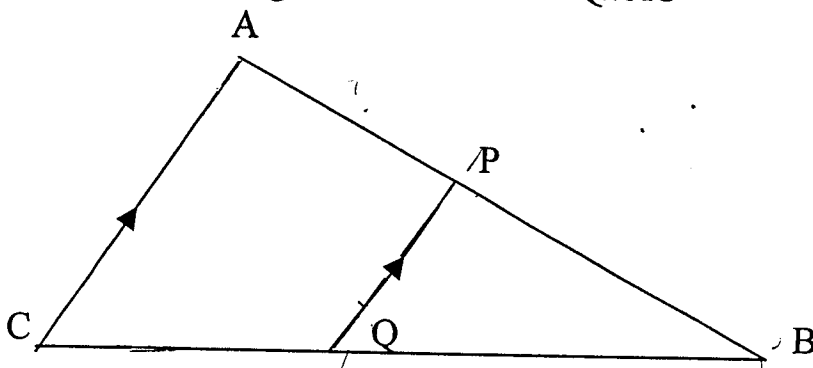
QUESTION 6.

- (a) One hundred tickets are sold in a raffle which has two prizes. First prize is drawn and the winning ticket discarded. Second prize is then drawn. Elizabeth buys 5 tickets in the raffle. What is the probability

- Elizabeth wins first prize?
- Elizabeth only wins first prize?
- Elizabeth only wins second prize?
- Elizabeth wins both prizes.

[4]

- (b) Given triangle ABC with line $PQ \parallel AC$

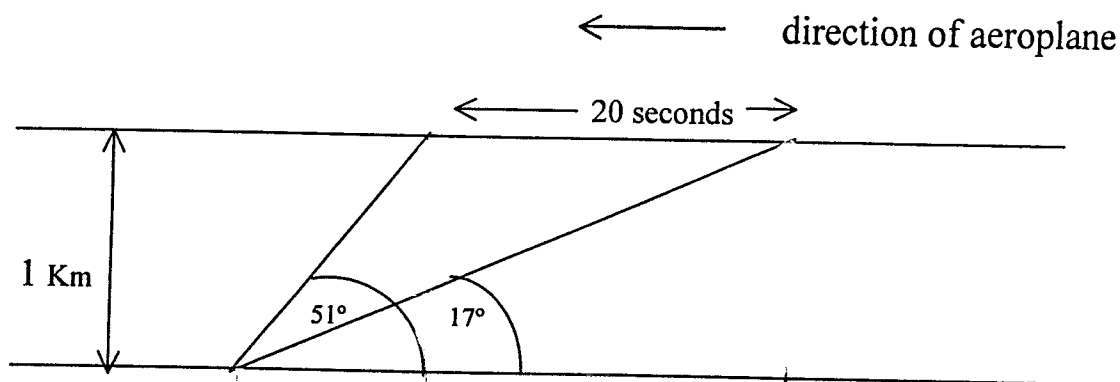


- Prove triangles ABC and PQB similar
- Find the length of AP given $PB = 8$, $QB = 10$ and $CQ = 4$

[5]

- (c) An aeroplane is flying at a level altitude of 1km above a flat plane. The angle of elevation of the aeroplane when it is first observed is 17° . Twenty seconds later the angle of elevation is 51° . What is the speed of the aeroplane in km/hr. (Correct to 3 significant figures)

[3]



QUESTION 7

- (a) Given the roots to the equation

$$2x^2 - 5x + 5 = 0 \text{ are } \alpha \text{ and } \beta$$

Write down the values of:

(i) $\alpha + \beta$

(ii) $\alpha \beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iv) $\alpha^2 + \beta^2$

[4]

- (b) A particle moves along the x – axis with acceleration at time t equal to $12(t + 3)^2$

i.e $\frac{d^2x}{dt^2} = 12(t + 3)^2$

If the particle is initially at rest at the origin (ie when $t=0$, $x=0$ and

$\frac{dx}{dt} = 0$), find the position of the particle when $t = 1$.

[4]

- (c) (i) Draw a neat sketch of the curves $y = x^2$ and $y = (x - 2)^2$
(ii) Find the co-ordinates of the point of intersection of the two curves
(iii) Find the area enclosed by the two curves and x -axis

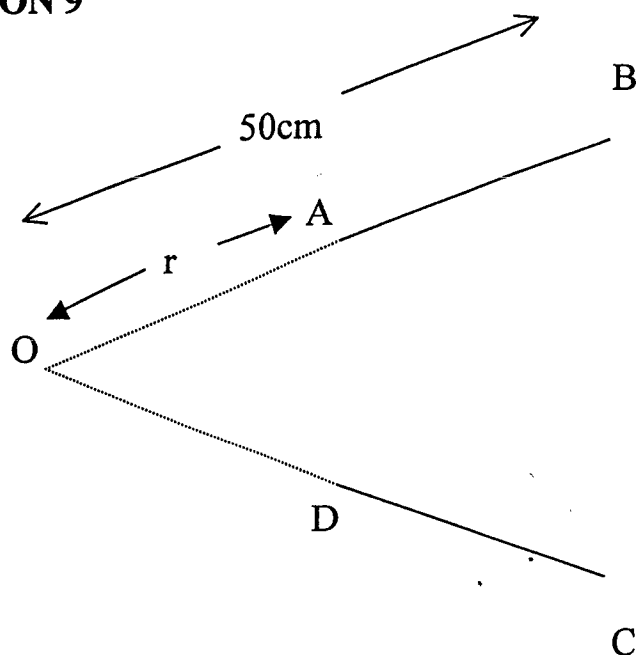
[4]

QUESTION 8

- (a) (i) Write down an expression for the sum of the series
 $1+3+5+\dots$ for n terms
- (ii) Write down an expression for the sum of the series
 $2+4+6+\dots$ for n terms
- (iii) Find the sum of the first 2000 terms of the series
 $1 - 2 + 3 - 4 + 5 - 6 + \dots$ [3]
- (b) Can there be an infinite geometric series with a limiting sum of
 $\frac{5}{8}$ and a first term of 2? (All working and reasoning must be shown) [2]
- (c) Indicate, by shading on a diagram, the region in the positive quadrant
 bounded by the y -axis, the line $y = 2$ and the curve $y = e^x$. Calculate
 the area of this region. [4]
- (d) The area under the curve $y = 4^x$ between $x = 0$ and $x = 2$ is rotated about
 the x -axis. Copy down and complete the table below. Use your
 results with Simpson's Rule to find an approximate value for the volume
 of rotation using 5 function values (correct to 1 d.p.) [3]
- (correct to 1 d.p.)

x	0	0.5	1	1.5	2
4^x					

QUESTION 9



OBC is a sector

AD and BC are arcs of concentric circles centre O

$\angle BOC = \theta$ (where θ is in radians)

$OA = r$ cm

$OB = 50$ cm

The perimeter of figure ABCD is 60cm

(i) Show that $\theta = \frac{2r - 40}{r + 50}$ [2]

(ii) Show that the area of ABCD is $= 70r - 1000 - r^2$ [2]

(iii) Find the values of r and θ which maximize the area of figure ABCD and find this maximum area [5]

- (b) Two people, John and Kim, play a dice game. John throws two dice and Kim throws one. Kim wins if John fails to beat his number with either one of his two dice. What is the probability that Kim wins the game? (Note: in the case of a draw Kim is declared the winner) [3]

QUESTION 10

- (a) A certain bacteria is growing according to the formula
 $B = B_0 e^{kt}$ where B_0 is the original population of bacteria
 t is time in hours
 k is a constant
And B_t the population of bacteria at time t

If the initial bacterial population is 10000 and doubles in size in 1 hour find:

- (i) the population of bacteria after $4\frac{1}{2}$ hours
- (ii) how long it will take, in hours, and minutes, for the population to reach 3 million

[4]

- (b) Jennifer borrowed \$100000 at 9% p.a reducible over 20 years to purchase a home unit. She repays \$M each month immediately after the monthly interest has been calculated and added to the loan.

- (i) Write down the balance owing at the end of the first month after interest has been added and a repayment deducted.
- (ii) Write down a formula for the balance owing after n repayments.
- (iii) Calculate the value of \$M, the monthly repayment, if the loan is repaid at the end of 20 years (answer to the nearest whole dollar).
- (iv) If Mary increases her repayment in part (iii) by \$100 from the first repayment, by how many months does she reduce the life of the loan? (answer to correct 1 decimal place)

[8]

Question One

a) 7.02

b) $\frac{2}{3-\sqrt{2}} = a+b\sqrt{2}$

$\frac{2}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$

$\frac{6+2\sqrt{2}}{9-2} = \frac{6}{7} + \frac{2}{7}\sqrt{2}$

$\therefore a = \frac{6}{7}, b = \frac{2}{7}$

c) $3x^2 - 2x - 5 = 0$

$(3x-5)(x+1) = 0$

$\therefore x = \frac{5}{3}, -1$

d) $|2x-3| \leq 7$

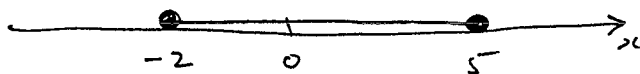
$2x-3 \leq 7 \quad -(2x-3) \leq 7$

$2x \leq 7+3 \quad -2x+3 \leq 7$

$x \leq 5 \quad -4 \leq 2x$

$-2 \leq x$

$\therefore -2 \leq x \leq 5$



e) 0.47

Let $x = 0.4777...$

$10x = 4.7777$

$9x = 4.3$

$x = \frac{43}{90}$

f)(i) $\frac{1}{p} = \frac{1}{q} + \frac{1}{r}$

$\frac{1}{p} - \frac{1}{r} = \frac{1}{q}$

$\frac{R-P}{PR} = \frac{1}{Q}$

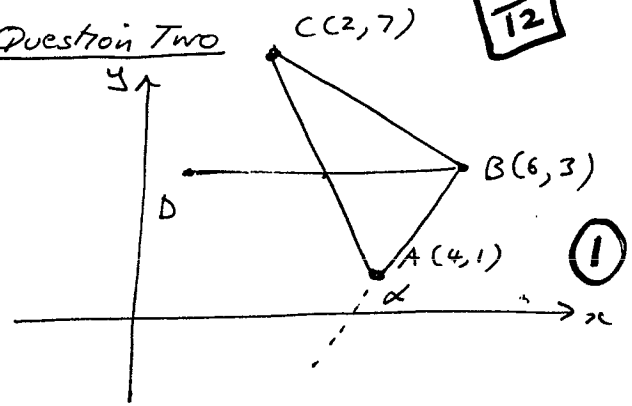
$Q = \frac{PR}{R-P}$

(ii) $Q = \frac{0.2 \times 0.5}{0.5 - 0.2}$

$Q = 0.3$

Question Two

a)



b) Grad of AB $\Rightarrow m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-1}{6-4} = 1$

c) Eqn of AB $y - y_1 = m(x - x_1)$

$y - 1 = 1(x - 4)$

$y = x - 3$

d) $\tan \alpha = 1$

$\therefore \alpha = \frac{\pi}{4}$ or $\alpha = 45^\circ$

e) Distance AB $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $= \sqrt{(6-4)^2 + (3-1)^2}$
 $= \sqrt{4+4} = \sqrt{8}$
 $= 2\sqrt{2}$ units

f) Grad of BC $m_2 = \frac{7-3}{2-6} = \frac{4}{-4} = -1$

Now $AB \perp BC$ if $m_1 \times m_2 = -1$

$\Rightarrow -1 \times 1 = -1$

$\therefore AB \perp BC$

i.e. $\triangle ABC$ is right angled.

g) mid-point of AC.

$x = \frac{2+4}{2} = \frac{6}{2} = 3$

$y = \frac{7+1}{2} = \frac{8}{2} = 4$

h) Same mid-point for BD as AC.

Co-ords of D. $x = 3 = \frac{x_2 + 6}{2}$

$\therefore x_2 = 0$
 $y = 4 = \frac{y_2 + 3}{2}$

$\therefore D(0,5)$

(i) Distance BC $= \sqrt{(6-2)^2 + (3-7)^2} = \sqrt{4^2 + 4^2} = \sqrt{32}$

Area of rectangle $= 2\sqrt{2} \times 4\sqrt{2} = 16$ s.u.

(j) $d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right| = \left| \frac{0+0-3}{\sqrt{1^2+1^2}} \right| = \left| \frac{-3}{\sqrt{2}} \right| = \frac{3}{\sqrt{2}}$ units

Question Three

12

a) (i) $y = 4x^3 - \frac{4}{x^3} = 4x^3 - 4x^{-3}$ ①
 $\frac{dy}{dx} = 12x^2 + 12x^{-4} = 12x^2 + \frac{12}{x^4}$

(ii) $y = \ln\left(\frac{x^2-5}{x+3}\right) = \ln(x^2-5) - \ln(x+3)$
 $\frac{dy}{dx} = \frac{2x}{x^2-5} - \frac{1}{x+3}$
 $= \frac{2x^2+6x-x^2-5}{(x^2-5)(x+3)} = \frac{x^2+6x-5}{(x^2-5)(x+3)}$ ②

(iii) $y = x^2 \cdot e^{\cos x}$
 $\frac{dy}{dx} = 2x \cdot e^{\cos x} + x^2 \cdot -\sin x \cdot e^{\cos x}$
 $= 2x \cdot e^{\cos x} - \sin x \cdot x^2 \cdot e^{\cos x}$ ②

b) (i) $\int \frac{dx}{3x-1} = \frac{1}{3} \int \frac{3 \cdot dx}{3x-1}$
 $= \frac{1}{3} \ln(3x-1) + C$ ①

(ii) $\int (2x+1)^9 \cdot dx$
 $= \frac{1}{10} \cdot \frac{1}{2} \cdot (2x+1)^{10} + C$
 $= \frac{1}{20} (2x+1)^{10} + C$ ①

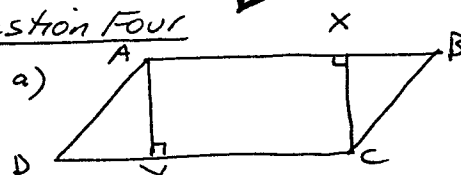
c) $\int_0^1 e^{4x} \cdot dx = \left[\frac{1}{4} e^{4x} \right]_0^1$
 $= \left[\frac{1}{4} e^4 - \frac{1}{4} e^0 \right]$
 $= \frac{1}{4} (e^4 - 1)$ ②

d) (i) $y = x \cdot \sin x + \cos x$
 $\frac{dy}{dx} = 1 \cdot \sin x + x \cdot \cos x - \sin x$
 $= x \cdot \cos x$ ①

(ii) $\int_{\frac{\pi}{2}}^{\pi} x \cdot \cos x \cdot dx$
 $= \left[x \cdot \sin x + \cos x \right]_{\frac{\pi}{2}}^{\pi}$ (from i)
 $= \left[\pi \cdot \sin \pi + \cos \pi \right] - \left[\frac{\pi}{2} \cdot \sin \frac{\pi}{2} + \cos \frac{\pi}{2} \right]$
 $= \left[\pi \cdot 0 - 1 \right] - \left[\frac{\pi}{2} \cdot 1 + 0 \right]$
 $= -1 - \frac{\pi}{2} = -\left(1 + \frac{\pi}{2}\right)$ ②

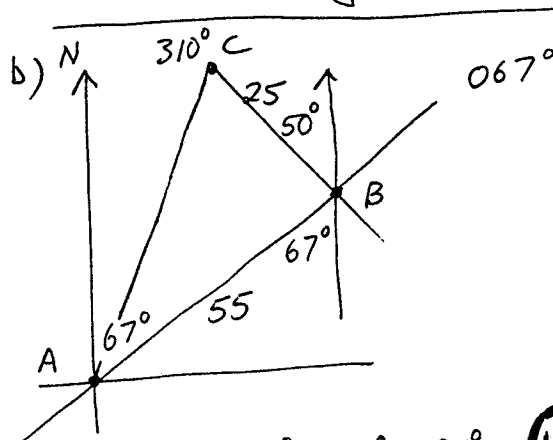
Question Four

12



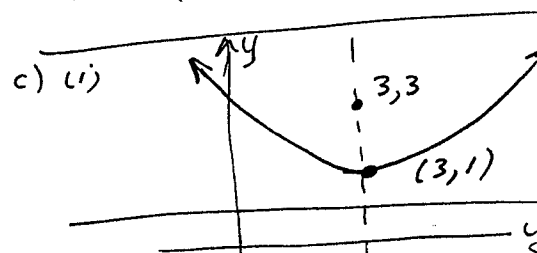
- a) (i) R.T.P. $\triangle ADY = \triangle CBX$
 1. $AD = BC$ (opp sides of rect equal)
 2. $\angle ADY = \angle CBX$ (opp $\angle$ s of rect equal)
 3. $\angle AYD = \angle CXB = 90^\circ$ (data)
 $\therefore \triangle ADY = \triangle CBX$ (AAS)

- (ii) $AY = CX$ (corres. sides of cong $\triangle$ s)



(i) $\angle ABC + 50^\circ + 67^\circ = 180^\circ$
 $\angle ABC = 63^\circ$

(ii) $AC^2 = 55^2 + 25^2 - 2 \times 55 \times 25 \times \cos 63^\circ$
 $AC = 49$



- (ii) axis of symmetry $x = 3$

(iii) Parabola with $a = 2$
 $(x-3)^2 = 8(y-1)$

(iv) $y = \frac{1}{8}(x^2 - 6x + 17)$
 $\frac{dy}{dx} = \frac{2x}{8} - \frac{6}{8}$
 At $x = 0$, $y = \frac{17}{8}$
 $\frac{dy}{dx} = -\frac{3}{4}$
 Eqn of tangent $y - \frac{17}{8} = -\frac{3}{4}(x - 0)$
 $y = -\frac{3}{4}x + \frac{17}{8}$ *

Question Five

a) $y = 3x^2 - x^3$

(i) $\frac{dy}{dx} = 6x - 3x^2$

St. pts $\frac{dy}{dx} = 0 \Rightarrow 6x - 3x^2 = 0$
 $3x(2-x) = 0$

$\therefore x = 0, x = 2 \quad (0, 0)$

$y = 0, y = 4 \quad (2, 4)$

$\frac{d^2y}{dx^2} = 6 - 6x$

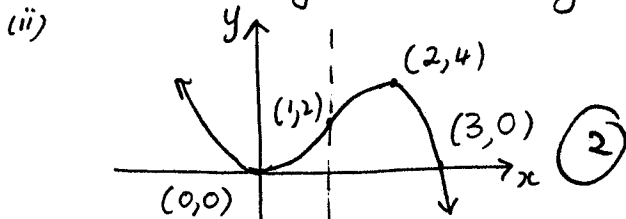
at $x = 0 \quad \frac{d^2y}{dx^2} > 0 \therefore \text{MIN}$

at $x = 2 \quad \frac{d^2y}{dx^2} < 0 \therefore \text{MAX}$

$(0, 0) \text{ MIN } (2, 4) \text{ MAX}$

$\frac{d^2y}{dx^2} = 6 - 6x = 0 \Rightarrow x = 1$
 $y = 2$

$(1, 2)$ change of concavity



(iii) at $(3, 0) \quad \frac{dy}{dx} = 18 - 27 = -9$

Grad of normal $= \frac{1}{9}$

Eqn of normal $y - 0 = \frac{1}{9}(x - 3)$
 $y = \frac{x}{9} - \frac{1}{3} *$

b) $T_3 + T_4 = 24 \quad T_5 + T_6 = 32$

$a + 2d + a + 3d = 24 \quad a + 4d + a + 5d = 32$

$2a + 5d = 24 \quad \text{--- (1)} \quad 2a + 9d = 32 \quad \text{--- (2)}$

(2) - (1) $\Rightarrow 4d = 8$
 $\underline{\underline{d = 2}} \Rightarrow \underline{\underline{a = 7}}$

(i) $7, 9, 11$

(ii) $= S_n = \frac{n}{2} \{2a + (n-1)d\}$

$S_{20} = \frac{20}{2} \{14 + (19) \times 2\}$
 $= 520$

c) $4x^2 + (1+m)x + 1 = 0$

$b^2 - 4ac = 0 \Rightarrow (m+1)^2 - 4 \cdot 4 \cdot 1 = 0$

$\therefore (m+1)^2 = 16$

$m+1 = \pm 4$

$\underline{\underline{m = 3, -5}}$

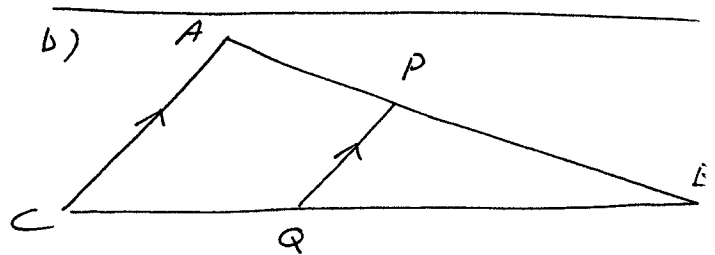
Question Six

a) (i) $P(\text{Ist}) = \frac{5}{100} = \frac{1}{20}$ (1)

(ii) $P(\text{Ist/Not Second}) = \frac{5}{100} \times \frac{95}{99}$
 $= \frac{19}{396}$ (1)

(iii) $P(\text{Not Ist/2nd}) = \frac{95}{100} \times \frac{5}{99} = \frac{19}{396}$ (1)

(iv) $P(\text{Ist/2nd}) = \frac{5}{100} \times \frac{4}{99} = \frac{1}{495}$ (1)



(i) R.T.P. $\triangle ABC \parallel \triangle PBQ$

1. $\angle CAB = \angle QPB$ (corres $\angle$'s $AC \parallel PQ$)

2. $\angle ACB = \angle PQB$ (.....)

3. $\angle B$ is common.

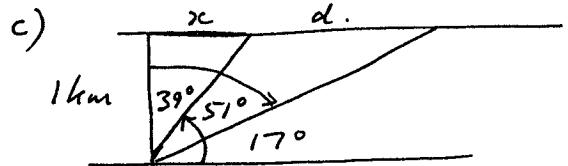
$\therefore \triangle ABC \parallel \triangle PBQ$ (equiangular).

(ii) Now: $\frac{AB}{PB} = \frac{CB}{QB}$ (corres sides of similar $\triangle$ s are in proportion)

$\therefore \frac{x+8}{8} = \frac{10+4}{10}$

$10x + 80 = 112 \Rightarrow 10x = 32$

$x = 3.2$



$\tan 39^\circ = \frac{x}{1 \text{ km}} \Rightarrow x = 0.81$

$\tan 73^\circ = \frac{x+d}{1 \text{ km}} \Rightarrow x+d = 3.2$

$\therefore d = 3.27 - 0.81 \text{ km}$

$d = 2.461 \text{ km in } 20 \text{ secs}$

Speed $= 443 \text{ km/hr}$

$$\tan 39^\circ = \frac{x}{1} = x.$$

$$\tan 73^\circ = \frac{A}{1} = A.$$

$$\therefore \tan 73^\circ - \tan 39^\circ = 2.46.$$

$$\therefore d = s \times t. \quad s = \frac{d}{t}.$$

$$s = \frac{2.46}{\frac{20/60}{60}} = 442.8$$

$$= 443 \text{ km/hr. } \checkmark$$

Q. SEVEN

$$7. a. \quad 2x^2 - 5x + 5 = 0 \quad \alpha, \beta.$$

$$i. \quad \alpha + \beta = \frac{-b}{a} = \frac{5}{2}. \quad \checkmark$$

$$ii. \quad \alpha \beta = \frac{c}{a} = \frac{5}{2}. \quad \checkmark$$

$$iii. \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha \beta} = \frac{5/2}{5/2} = 1. \quad \checkmark$$

$$iv. \quad \alpha^2 + \beta^2 = (\alpha^2 + \beta^2 + 2\alpha\beta) - 2\alpha\beta \\ = (\alpha + \beta)^2 - 2\alpha\beta \\ = \frac{25}{4} - \frac{10}{2} = \frac{5}{4}. \quad \checkmark$$

$$b. \quad \frac{d^2x}{dt^2} = 12(t+3)^2$$

$$\frac{dx}{dt} = \frac{12(t+3)^3}{3} = 4(t+3)^3 + c.$$

$$\text{when } \frac{dx}{dt} = 0 \quad t=0 \quad \therefore 0 = 27 \times 4 + c \\ c = -108. \quad \checkmark$$

$$\frac{dx}{dt} = 4(t+3)^3 - 108.$$

$$x = \frac{4(t+3)^4}{4} - 108t + c.$$

$$x = (t+3)^4 - 108t + c.$$

$$\text{when } x=0, \quad t=0.$$

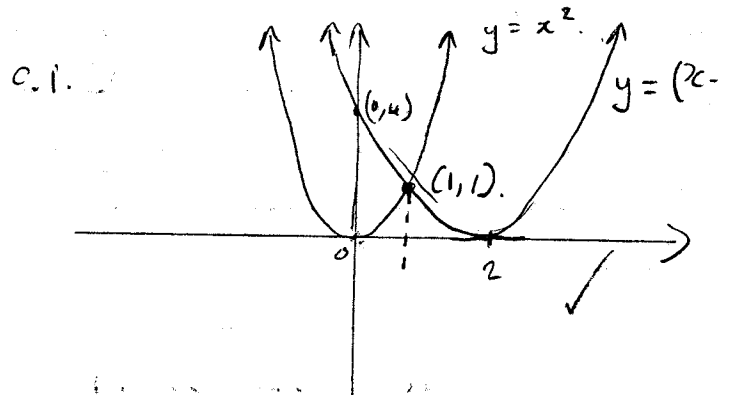
$$0 = 3^4 + c$$

$$c = -81 \quad \checkmark$$

$$\therefore x = (t+3)^4 - 108t - 81.$$

$$\text{when } t=1.$$

$$x = 4^4 - 108 - 81 \\ = 67. \quad \checkmark$$



$$\therefore y = x^2, \quad y = (x-2)^2.$$

$$x^2 = (x-2)^2$$

$$x^2 = x^2 - 4x + 4.$$

$$4x - 4 = 0$$

$$4x = 4.$$

$$x = 1. \quad \checkmark$$

$$\therefore y = 1$$

$$iii. \quad \int_0^1 x^2 - (x-2)^2. \quad \int_0^1 x^4 dx + \\ \int_0^1 x^2 - (x^2 - 4x + 4). \quad \int_1^2 (x-2)^2 dx \\ \int_0^1 4x - 4. \quad = \left[\frac{x^3}{3} \right]_0^1 + \left[\frac{(x-2)^3}{3} \right]_1^2 \\ = \left[\frac{4x^2}{2} - 4x \right]_0^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \text{ units} \\ = [2x^2 - 4x]_0^1 \\ = [2 - 4] - [0] \\ = -2$$

$\therefore \text{Area} = 2$ since it cannot be a negative no.

QUESTION EIGHT

$$\begin{aligned} 8. a.i. S_n &= \frac{n}{2} (2a + (n-1)d) \\ &= \frac{n}{2} (2 + (n-1)2) \\ &= \frac{n}{2} (2 + 2n - 2) = n^2 \checkmark \end{aligned}$$

$$\begin{aligned} ii. S_n &= \frac{n}{2} (2a + (n-1)d) \\ S_n &= \frac{n}{2} (4 + (n-1)2) \\ &= \frac{n}{2} (4 + 2n - 2) \\ &= \frac{n}{2} (2 + 2n) \\ &= n + n^2 \checkmark \\ &= n^2 + n \end{aligned}$$

6

$$\begin{aligned} iii. S_{1000} &= S_{1000}^{(odd)} - (S_{1000}^{(Even)}) \\ &= n^2 + n - (n^2) = n. \\ n &= 1000. \\ &= n^2 - (n^2 + n) \\ &= -n = -1000 \end{aligned}$$

$$b. S_{\infty} = \frac{a}{1-r} \quad |r| < 1$$

$$\frac{5}{8} = \frac{2}{1-r}$$

$$\text{or } \frac{5}{8} = \frac{2}{r-1}$$

$$5 - 5r = 16$$

$$5r - 5 = 16$$

$$5r = -11$$

$$5r = 21$$

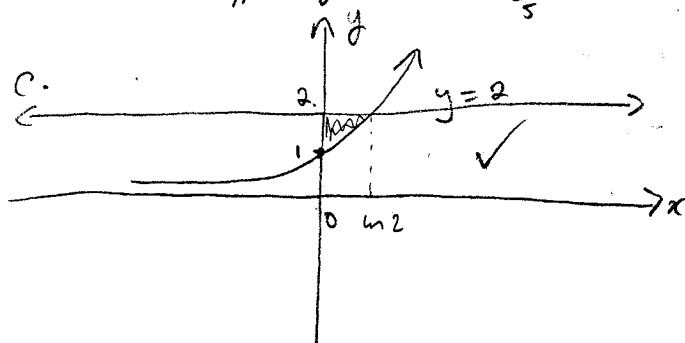
$$r = -\frac{11}{5}$$

$$r = \frac{21}{5}$$

$$r = -\frac{11}{5} \quad \text{since } |r| < 1$$

$$r > 1$$

No there is no limiting sum since $|r| > 1$: $\frac{21}{5} > 1$.



$$\begin{aligned} 2 &= e^x \quad \therefore \int_0^{\ln 2} 2 - e^x \\ \ln 2 &= x \quad = \left[2x - \frac{1}{2} e^x \right]_0^{\ln 2} \\ &= \left[2\ln 2 - \frac{2}{\ln 2} \right] = \left[2\ln 2 - e^{\ln 2} \right] - \left[\right] \\ &= 2\ln 2 - 2 + 1 \\ &= \ln 2^2 - \frac{2}{\ln 2} \quad \text{units}^2 = 2\ln 2 - 1 \end{aligned}$$

$$d. y = 4^x \quad x=0, x=2$$

x	0	0.5	1	1.5	2
4 ^x	1	2	4	8	16
(4 ^x) ²	1	4	16	64	256

$$A \doteq \frac{h}{3} (\text{sum end} + 4 \times \text{first} + 2 \times \text{second ordinates})$$

$$\doteq \frac{1/2}{3} (17 + 4(2) + 2(4) + 4(8))$$

$$\doteq \frac{1}{6} (17 + 48) \quad \text{Vol} = \pi \int_0^2 y^2 dx$$

$$\doteq 10.8 \quad = \pi \int_0^2 (4^x)^2 dx = 293.7 \text{ m}^3$$

$$9. i. \text{ Show } \theta = \frac{2r-40}{r+50}$$

$$L \doteq r\theta$$

$$\therefore \text{Perimeter.}$$

$$BC = 50\theta$$

$$2(50-r) + 50\theta + r\theta =$$

$$AD = r\theta$$

$$100 - 2r + \theta(50+r) =$$

$$\theta(50+r) = 2r-40$$

$$\theta = \frac{2r-40}{r+50} \checkmark$$

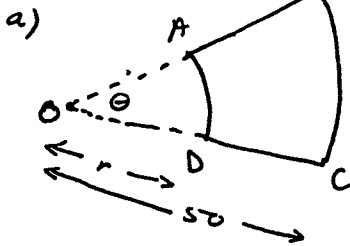
$$* ii. \text{ Area of segment ABCD.}$$

$$= \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times$$

Question Nine

(12)



$$\begin{aligned} (i) \quad AB &= 50 - r \\ CD &= 50 - r \\ BC &= 50\theta \\ AD &= r\theta. \end{aligned}$$

$$\text{Perimeter} = 2(50 - r) + \theta(r + 50).$$

$$\therefore 60 = 100 - 2r + \theta(r + 50)$$

$$2r - 40 = \theta(r + 50)$$

$$* \quad \theta = \frac{2r - 40}{r + 50} \quad (2)$$

(ii) Area of ABCD

$$= \frac{1}{2} \theta (50^2 - r^2)$$

$$A = \frac{1}{2} \cdot \frac{2r - 40}{r + 50} (50 - r)(50 + r)$$

$$A = (r - 20)(50 - r)$$

$$* \quad A = 70r - 1000 - r^2 \quad (2)$$

(iii) $A = 70r - 1000 - r^2$

$$\frac{dA}{dr} = 70 - 2r \quad (\text{For Max/min})$$

$$\therefore 70 - 2r = 0$$

$$\Rightarrow r = 35.$$

$$\frac{d^2A}{dr^2} = -2 < 0 \therefore \text{only max possible.}$$

Max area when $r = 35$

$$\text{Now } \theta = \frac{2 \times 35 - 40}{35 + 50} = \frac{30}{85}$$

$$\theta = \frac{6}{17} \text{ (radians)}$$

Now Maximum Area

$$A = \frac{1}{2} \cdot \frac{6}{17} (50^2 - 35^2)$$

$$A = 225 \text{ cm}^2 \quad (5)$$

(b) Kim wins with a draw or larger

K	J ₁	J ₂	
1	1	1	1
2	2	2	4
3	3	3	9
4	4	4	16
5	5	5	25
6	6	6	36

$$\text{Kim Wins} = \frac{91}{216}$$

(3)

Question Ten

(12)

$$a) \quad B = B_0 e^{kt}$$

$$B_0 = 10000$$

$$20000 = 10000 e^k$$

$$B = 20000 \quad t =$$

$$2 = e^k$$

$$k = \ln 2$$

$$(i) \quad B = 10000 e^{t \ln 2} \quad t = 4 \frac{1}{2}$$

$$\therefore B = 10000 e^{4 \frac{1}{2} \ln 2} =$$

$$B = 226274.17 \quad (2)$$

$$(ii) \quad 3000000 = 10000 e^{t \ln 2}$$

$$300 = e^{t \ln 2}$$

$$\therefore t = 8 \text{ hours } 14 \text{ minutes} \quad (2)$$

b) i) Balance after 1 month.

$$B = \$100000 \times (1.0075) - \$M$$

(2)

$$(ii) \quad \text{Balance} = \$100000 \times (1.0075)^n - \$M \left(\frac{1.0075^n - 1}{1.0075 - 1} \right)$$

(iii) for $n = 20$ Balance = 0.

$$\$100000 \times (1.0075)^{240} = \$M \left(\frac{1.0075^{240} - 1}{1.0075 - 1} \right)$$

$$\$M = \$899.73$$

$$\$M = \$900 \quad (3)$$

(iv) Require n if $\$M = \1000

$$\therefore \$100000 \times (1.0075)^n = \$1000 \left(\frac{1.0075^n - 1}{0.0075} \right)$$

$$100(1.0075)^n = \frac{(1.0075)^n - 1}{0.0075}$$

$$0.75(1.0075)^n = (1.0075)^n - 1$$

$$1 = (1.0075)^n (1 - 0.75)$$

$$1 = (1.0075)^n (0.25)$$

$$\therefore (1.0075)^n = 4$$

$$n \ln(1.0075) = \ln 4 \quad (2)$$

$$n = \frac{\ln 4}{\ln(1.0075)} = 185.5$$

$$\text{Months saved} = 240 - 185.5 = 55.5 \text{ months}$$